

Adam Rychtecky

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PROFESSIONAL SUMMARY

Data scientist with hands-on experience in machine learning, large-scale data pipelines, and quantitative research. Built and evaluated production-scale ML models across distributed systems and human-subjects experiments. Strong statistical foundation across regression, classification, clustering, and Bayesian methods, with the ability to communicate complex findings to technical and non-technical audiences.

EDUCATION

University of Washington (UW)

Master of Science in Data Science (M.S.).

March 2028

University of California, Santa Barbara (UCSB)

Bachelor of Science (B.S.) -- Statistics and Data Science; Psychological and Brain Science.

June 2026

EXPERIENCE

Executive Research Assistant

January 2024 – April 2026

Vision and Image Understanding Lab / Bionic Vision Lab, UC Santa Barbara

Goleta, CA

- Coordinated and administered psychophysics sessions for a deep-learning study predicting functional visual performance in bionic-eye users (Skaza et al., 2025 preprint).
- Supported a scene-understanding study using gaze-contingent simulated retinal prosthetic vision with graduate researcher Byron Johnson, running experimental sessions and eye-tracking calibration; work builds on Johnson et al., Journal of Vision, 2024.
- Implemented gaze-contingent and phosphene-rendered stimuli, ensuring reliable eye-tracking data collection across both labs.
- Organized and quality-checked datasets for statistical and machine learning analysis.
- Conducted large-scale human psychophysics experiments, including AI-human decision-making comparisons in visual perception tasks.

IT Intern

July 2024 – August 2024

UFCW Trust

Concord, CA

- Oversaw the upgrade of 150+ company laptops through Windows Autopilot, installing software personalized to each employee.
- Collaborated with colleagues to develop a project plan that optimized workflow and shortened completion time by 3 weeks.
- Documented serial numbers for new and retiring hardware, ensuring proper data wipe per HIPAA guidelines.

PROJECTS

Flexible Wisdom: Human vs. LLM Ensemble Decision-Making (Honors Thesis)

- Designed and ran a controlled comparison of human and LLM ensemble decision-making on a two-alternative forced-choice visual detection task, benchmarked against a Bayesian ideal observer.
- Found that LLM ensembles approached the Bayesian ideal within 3 to 5 percentage points, while human ensembles stayed 13 to 25 points below it across conditions.
- Showed that accuracy gains saturate around five agents due to positive error correlation between agents (mean correlation ~0.29).
- Built the analysis pipeline in Python across 500k+ trials to quantify aggregation behavior and failure modes in LLM systems.

Court-Vision: Distributed Analysis of 4.2M+ NBA Shot Records

- Analyzed 4.2M+ NBA shot records from 2004 to 2024 on a distributed Apache Spark (PySpark) cluster on Google Cloud Dataproc, with data staged in Google Cloud Storage.
- Built a LASSO-regularized shot-make classifier with a full regularization path and ran zone-efficiency analysis across a 21-season series.
- Applied PCA and K-Means clustering to group players into shooting archetypes from their shot profiles.

SKILLS

Languages and Tools: Python, SQL, R, PySpark, Git

Machine Learning: Regression and classification, Regularization (LASSO), PCA, K-Means clustering, Time series, Generalized additive models, Tree-based models, LLM evaluation and ensemble analysis

Big Data and Cloud: Apache Spark, Google Cloud Dataproc, Google Cloud Storage, Distributed model training on multi-million-record datasets

Statistics and Research: Experimental design, Bayesian benchmarking, Hypothesis testing, Psychophysics, Data collection and quality control, Scientific writing